INDIVIDUAL CONTRIBUTION RECORD

1. Student Details

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PAP Group

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Module Developed: Reports Module

2. Overview of My Contribution

I was responsible for developing the Reports Module for the Municipal Financial Management System (MFMS).

The purpose of my module is to provide a central Reports menu through which users can access reports relating to the different sections of the system. The module provides reports for Employees, Budget, Suppliers and Assets.

I developed the Reports module using C and separated the declarations and implementation into a header file and source file. This follows the modular structure used by the rest of the project.

3. Files Developed

reports.h

The reports.h file contains the function declarations for the Reports module. The functions declared include:

employeeReport()

budgetReport()

supplierReport()

assetReport()

reportsMenu()

The header file allows the Reports functions to be accessed by other parts of the system when the module is integrated.

reports.c

The reports.c file contains the implementation of the Reports module. It includes the required header files and contains the report functions and Reports menu.

4. Functions Implemented

employeeReport()

This function provides the Employee Report. It calls the existing Employee Management display function so that the employee information stored in the system can be displayed as part of the report.

budgetReport()

This function provides the Budget Report. It uses the existing Budget Management functions to display the recorded budget information and departments that are over budget.

supplierReport()

This function provides the Supplier Report. It uses the existing Supplier Management display function to display the registered suppliers and their available information.

assetReport()

This function provides the Asset Report. It uses the existing Asset Management display function to display the assets registered in the system.

reportsMenu()

This function provides the user with a menu for selecting the required report. A switch statement is used to determine which report function should be executed.

5. Data Structures and Programming Concepts

The Reports module makes use of the data already maintained by the individual system modules rather than creating separate copies of the information.

The following programming concepts were applied:

Functions – separate functions were created for each report.

Modular programming – the Reports functionality was separated into reports.c and reports.h.

Header files – reports.h contains the declarations of the Reports functions.

Switch statements – used in the Reports menu to process the user's selection.

Do-while loop – used to keep the Reports menu active until the user chooses to return to the Main Menu.

Function calls – the Reports functions call the existing functions of the Employee, Budget, Supplier and Asset modules.

Input handling – scanf() and getchar() are used to receive and process menu input.

6. GitHub Contribution

I created the Reports module files and committed my work to the group's GitHub repository.

My contribution included:

Creating reports.h.

Creating reports.c.

Adding the Reports function declarations.

Implementing the Reports menu.

Implementing the four report functions.

Committing the changes to GitHub.

Pushing my contribution to the group repository.

The GitHub commit history provides evidence of my contribution to the project.

7. Contribution to the Overall System

The Reports Module provides a central location for accessing information from the different management sections of the MFMS.

Instead of requiring the user to navigate through each management section separately, the Reports menu provides direct access to the Employee, Budget, Supplier and Asset reports.

8. Conclusion

My contribution to the project was the development of the Reports Module. I created the required header and source files, implemented the report functions and created the Reports menu.

Through this contribution, I applied concepts such as functions, modular programming, header files, loops, switch statements and function calls. I also gained experience working with a shared GitHub repository and contributing code as part of a group programming project.